BTEC Level 3 Applied Science · Unit 1 Physics · Answer sheet

Wave Properties — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording. These are original JDScience items, not official Pearson questions.

Numbering matches the student worksheet exactly. Award equivalent scientifically correct wording. Show units on every calculated quantity.

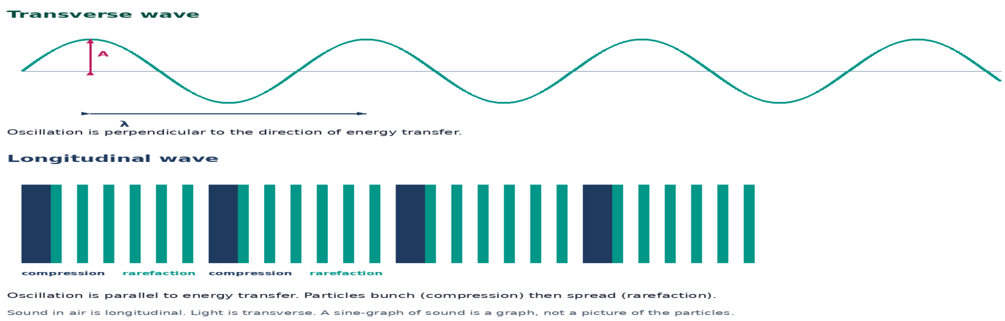
A–E

1. Transverse: oscillation perpendicular to energy transfer. Longitudinal: parallel.

2. Light / water / string; sound / P-waves / slinky compressions.

3. Resultant displacement is the sum of individual displacements.

4. Compression: particles closer / higher pressure. Rarefaction: more spaced / lower pressure.



5. They reach maxima and minima together / separated by nλ.

6. Light is transverse so oscillations have a direction that can be filtered; sound is longitudinal.

7. Ultrasound longitudinal; microwaves transverse; stadium sound longitudinal.

8. Yes — separation is 3λ, a whole number of wavelengths.

9. See definitions plus one valid example of each. [4]

10. Resultant displacement = sum of individual displacements. [2]

11. Polarisation needs a transverse direction; sound is longitudinal. [2]

12. The sine curve is a graph of pressure or displacement. The air particles still oscillate parallel to the direction of travel.